

Forecasting

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```
bookdown::render_book()
```


Overview

1. Preparing for Forecasting
2. Forecasting and Comparing Models

Chapter 1

Prepare for Forecasting

1.1 Graph the Time Series

We'll use our Phillip's Curve data to forecast unemployment one-step into the future

Set Time Series

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use phillips.dta, clear  
tsset year
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

```
time variable: year, 1948 to 2003  
delta: 1 unit
```

Use *tsline* command to graph the time series

```
tsline inf  
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_11_inflation.png", replace
```

1.2 Train and Test

First we need to Generate training and test samples, so we'll use the time series sequence until 1996 as our training data, and we'll use 1997 to 2003 as our testing data.

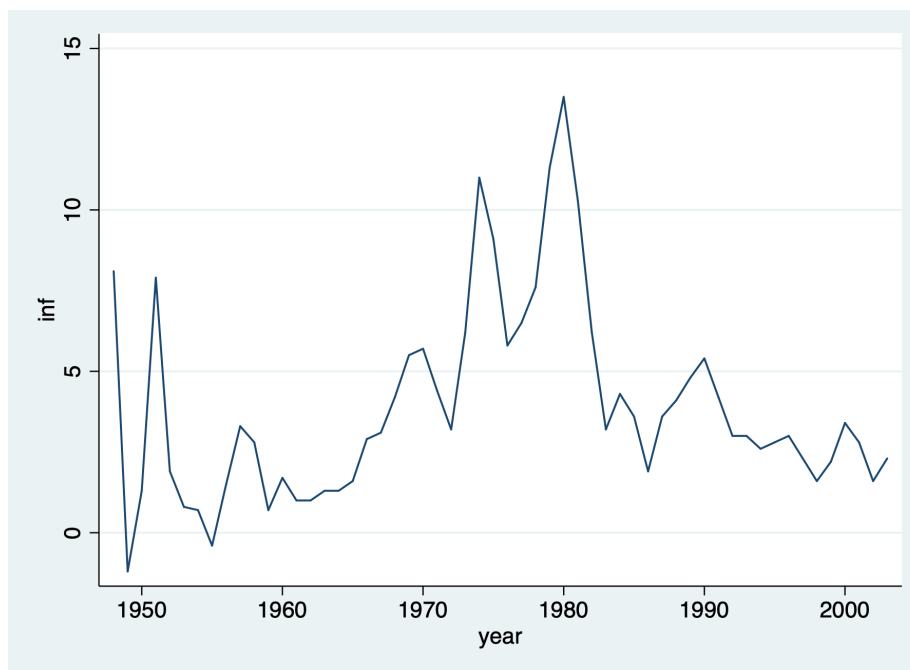


Figure 1.1: Graph of Inflation Rate

```

gen test = 0
replace test = 1 if year >= 1997
label define test1 0 "training sample" 1 "testing sample"
label values test test1
list year unem inf test

```

(7 real changes made)

	year	unem	inf	test
1.	1948	3.8	8.1000004	training sample
2.	1949	5.9000001	-1.2	training sample
3.	1950	5.3000002	1.3	training sample
4.	1951	3.3	7.9000001	training sample
5.	1952	3	1.9	training sample
6.	1953	2.9000001	.80000001	training sample
7.	1954	5.5	.69999999	training sample
8.	1955	4.4000001	-.40000001	training sample

9.		1956	4.0999999	1.5	training sample	
10.		1957	4.3000002	3.3	training sample	

11.		1958	6.8000002	2.8	training sample	
12.		1959	5.5	.69999999	training sample	
13.		1960	5.5	1.7	training sample	
14.		1961	6.6999998	1	training sample	
15.		1962	5.5	1	training sample	

16.		1963	5.6999998	1.3	training sample	
17.		1964	5.1999998	1.3	training sample	
18.		1965	4.5	1.6	training sample	
19.		1966	3.8	2.9000001	training sample	
20.		1967	3.8	3.0999999	training sample	

21.		1968	3.5999999	4.1999998	training sample	
22.		1969	3.5	5.5	training sample	
23.		1970	4.9000001	5.6999998	training sample	
24.		1971	5.9000001	4.4000001	training sample	
25.		1972	5.5999999	3.2	training sample	

26.		1973	4.9000001	6.1999998	training sample	
27.		1974	5.5999999	11	training sample	
28.		1975	8.5	9.1000004	training sample	
29.		1976	7.6999998	5.8000002	training sample	
30.		1977	7.0999999	6.5	training sample	

31.		1978	6.0999999	7.5999999	training sample	
32.		1979	5.8000002	11.3	training sample	
33.		1980	7.0999999	13.5	training sample	
34.		1981	7.5999999	10.3	training sample	
35.		1982	9.6999998	6.1999998	training sample	

36.		1983	9.6000004	3.2	training sample	
37.		1984	7.5	4.3000002	training sample	
38.		1985	7.1999998	3.5999999	training sample	
39.		1986	7	1.9	training sample	
40.		1987	6.1999998	3.5999999	training sample	

41.		1988	5.5	4.0999999	training sample	
42.		1989	5.3000002	4.8000002	training sample	
43.		1990	5.5999999	5.4000001	training sample	
44.		1991	6.8000002	4.1999998	training sample	
45.		1992	7.5	3	training sample	

46.		1993	6.9000001	3	training sample	

47.		1994	6.0999999	2.5999999	training sample	
48.		1995	5.5999999	2.8	training sample	
49.		1996	5.4000001	3	training sample	
50.		1997	4.9000001	2.3	testing sample	

51.		1998	4.5	1.6	testing sample	
52.		1999	4.1999998	2.2	testing sample	
53.		2000	4	3.4000001	testing sample	
54.		2001	4.8000002	2.8	testing sample	
55.		2002	5.8000002	1.6	testing sample	

56.		2003	6	2.3	testing sample	

1.3 AR(1) and VAR models

We'll use two models to forecast unemployment:

1. AR(1) model: Just a 1-year lag of unemployment
2. VAR model: 1-year lag of unemployment and 1-year lag of inflation

Run the regressions using training data

OLS AR-1

Run the model with a 1-year lag in unemployment

```
reg unem 1.unem if test == 0
```

Source	SS	df	MS	Number of obs	=	48
				F(1, 46)	=	57.13
Model	62.8162728	1	62.8162728	Prob > F	=	0.0000
Residual	50.5768515	46	1.09949677	R-squared	=	0.5540
				Adj R-squared	=	0.5443
Total	113.393124	47	2.41261967	Root MSE	=	1.0486

unem	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
unem					
L1.	.7323538	.0968906	7.56	0.000	.537323 .9273845
_cons	1.571741	.5771181	2.72	0.009	.4100629 2.73342

VAR

We will add a 1-year lag in inflation along with our 1-year lag in unemployment

```
reg unem l.unem l.inf if test == 0
```

Source	SS	df	MS	Number of obs	=	48
				F(2, 45)	=	50.22
Model	78.3083336	2	39.1541668	Prob > F	=	0.0000
Residual	35.0847907	45	.779662015	R-squared	=	0.6906
				Adj R-squared	=	0.6768
Total	113.393124	47	2.41261967	Root MSE	=	.88298

unem	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
unem					
L1.	.6470261	.0838056	7.72	0.000	.4782329 .8158192
inf					
L1.	.1835766	.0411828	4.46	0.000	.1006302 .2665231
_cons	1.303797	.4896861	2.66	0.011	.3175188 2.290076

Chapter 2

Forecasting

2.1 One-Step Ahead Model vs Dynamic AR(1) Model

One-step ahead we can use predict or forecast for AR(1). We use the *estimates store* command to store our results for our AR(1) model.

Train the model

```
reg unem l.unem if test == 0
estimates store model1
```

One-Step Ahead Model

We can get one-step ahead just using predict

```
predict unem_est
```

Forecast - One-step-ahead forecast. We three addition commands:

1. *forecast create*
2. *forecast estimates*
3. *forecast solve*

```
forecast create forecast1, replace
forecast estimates model1
```

For a one-step ahead forecast, **static** is our key option otherwise it will not be a one-step ahead forecast.

```
forecast solve, prefix(st_) begin(1997) end(2003) static
list year unem st_unem if year > 1996
```

Computing static forecasts for model forecast1.

```
-----
Starting period: 1997
Ending period: 2003
Forecast prefix: st_
```

```
1997: .....
1998: .....
1999: .....
2000: .....
2001: .....
2002: .....
2003: .....
```

Forecast 1 variable spanning 7 periods.

```
-----
+-----+
| year      unem      st_unem |
+-----+
50. | 1997    4.9000001  5.5264521 |
51. | 1998           4.5   5.160275 |
52. | 1999    4.1999998  4.8673334 |
53. | 2000           4     4.6476274 |
54. | 2001    4.8000002  4.5011568 |
+-----+
55. | 2002    5.8000002  5.0870399 |
56. | 2003           6     5.8193936 |
+-----+
```

Dynamic Model

Dynamic model - We won't use l.unem but l.dy_unem

```
forecast solve, prefix(dy_) begin(1997) end(2003)
list year unem st_unem dy_unem if year > 1996
```

Computing dynamic forecasts for model forecast1.

```
-----
Starting period: 1997
Ending period:   2003
Forecast prefix: dy_
```

```
1997: .....
1998: .....
1999: .....
2000: .....
2001: .....
2002: .....
2003: .....
```

Forecast 1 variable spanning 7 periods.

```
-----+-----+
| year      unem      st_unem      dy_unem |
|-----+-----+
50. | 1997  4.9000001  5.5264521  5.5264521 |
51. | 1998      4.5    5.160275   5.6190596 |
52. | 1999  4.1999998  4.8673334  5.6868811 |
53. | 2000      4     4.6476274  5.7365503 |
54. | 2001  4.8000002  4.5011568  5.7729259 |
|-----+-----+
55. | 2002  5.8000002  5.0870399  5.7995653 |
56. | 2003      6     5.8193936  5.8190751 |
+-----+-----+
```

Graph Model

Graph Actual, One-Step-Ahead Model, and Dynamic Model

```
twoway line unem year || line st_unem year, lpattern(dash) || line dy_unem year, lpattern(dot)
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_11_static_v_dynamic.png", replace
```

Root Mean Squared Error (RMSE)

Calculate RMSE for AR(1) model

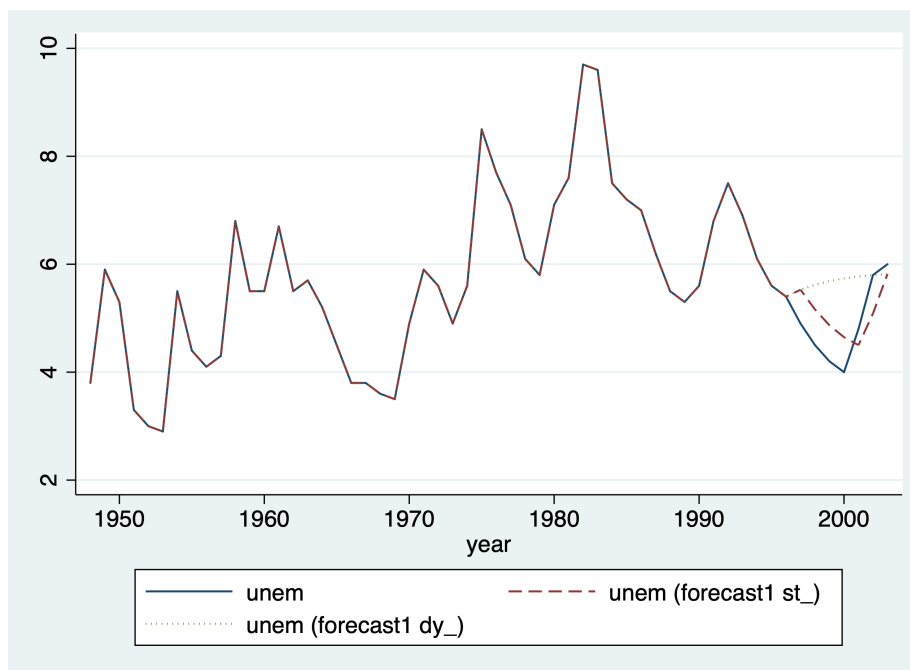


Figure 2.1: One-Step-Ahead vs Dynamic Forecasting

```

gen e = unem - st_unem
gen e2 = e^2
sum e2 if test==1
scalar esum= `r(sum)'/`r(N)'
scalar model1_RMSE= esum^.5
display model1_RMSE

```

Variable	Obs	Mean	Std. Dev.	Min	Max
e2	7	.3319141	.1891047	.0326187	.5083123

.57611988

Mean Absolute Error (MAE)

Calculate MAE for AR(1) model


```
gen e_abs = abs(e) if test ==1
sum e_abs if test==1
scalar model1_MAE=`r(mean)'
display model1_MAE
```

(49 missing values generated)

Variable	Obs	Mean	Std. Dev.	Min	Max
e_abs	7	.542014	.2109284	.1806064	.7129602

.54201399

2.2 One-Step Ahead VAR Model

One-step ahead we can use predict or forecast for VAR Model (lagged employment and lagged inflation)

Train the Model

```
quietly reg unem l.unem l.inf if year < 1997
```

Our predict command will produce the same results as *forecast solve, static*

```
predict p2
estimates store model2
```

Forecast for Model 2

```
forecast create forecast2, replace
forecast estimates model2
```

One-Step-Ahead Forecast of VAR Model

```
forecast solve, prefix(st2_) begin(1997) end(2003) static
list year unem st_unem if year > 1996
```

Computing static forecasts for model forecast2.

 Starting period: 1997
 Ending period: 2003
 Forecast prefix: st2_

1997:
 1998:
 1999:
 2000:
 2001:
 2002:
 2003:

Forecast 1 variable spanning 7 periods.

 +-----+
year unem st2_unem
 50. | 1997 4.9000001 5.3484678 |
 51. | 1998 4.5 4.896451 |
 52. | 1999 4.1999998 4.5091372 |
 53. | 2000 4 4.4251752 |
 54. | 2001 4.8000002 4.5160618 |
 |-----|
 55. | 2002 5.8000002 4.9235368 |
 56. | 2003 6 5.3502712 |
 +-----+

Dynamic Forecast of VAR Model

```
forecast solve, prefix(dy2_) begin(1997) end(2003)
list year unem st2_unem dy2_unem if year > 1996
```

Computing dynamic forecasts for model forecast2.

 Starting period: 1997
 Ending period: 2003
 Forecast prefix: dy2_

1997:
 1998:

```

1999: .....
2000: .....
2001: .....
2002: .....
2003: .....

```

Forecast 1 variable spanning 7 periods.

	year	unem	st2_unem	dy2_unem
50.	1997	4.9000001	5.3484678	5.3484678
51.	1998	4.5	4.896451	5.1866217
52.	1999	4.1999998	4.5091372	4.9533992
53.	2000	4	4.4251752	4.9126439
54.	2001	4.8000002	4.5160618	5.1065664
55.	2002	5.8000002	4.9235368	5.1218934
56.	2003	6	5.3502712	4.9115181

2.2.1 Calculate RMSE and MAE

Calculate RMSE for the VAR model

```

gen e = unem - st2_unem
gen e2 = e^2
sum e2 if test==1
scalar esum2 = `r(sum)'/`r(N)'
scalar model2_RMSE=esum2^.5
display model2_RMSE

```

Variable	Obs	Mean	Std. Dev.	Min	Max
e2	7	.2722276	.2459784	.080621	.7681881

.52175434

Calculate MAE for VAR model

```

gen e_abs = abs(e)
sum e_abs if test==1
scalar model2_MAE=`r(mean)'
display model2_MAE

```

Variable	Obs	Mean	Std. Dev.	Min	Max
e_abs	7	.4841946	.2099534	.2839384	.8764634

```
.48419455
```

2.3 Compare AR(1) and VAR models

Which model do we use? The model that minimizes the error.

Compare Results with RMSE and MSE

```

display "Model 1: RMSE="model1_RMSE " and MAE="model1_MAE
display "Model 2: RMSE="model2_RMSE " and MAE="model1_MAE
display "Model 2 with lagged unemployment and lagged inflation has lower RMSE and MAE.

```

```
Model 1: RMSE=.57611988 and MAE=.54201399
```

```
Model 2: RMSE=.52175434 and MAE=.48419455
```

```
Model 2 with lagged unemployment and lagged inflation has lower RMSE and MAE.
```

Graph the Forecasts

Forecasting Line

```

twoway line unem year || line st_unem year, lpattern(dash) || line st2_unem year, l
title("Forecast of Unemployment") xtitle("Year") subtitle("One-Step-Ahead") ytitle
legend(order(1 "Actual" 2 "Model 1" 3 "Model 2"))
graph export "/Users/Sam/Desktop/Econ 645/Stata/week_11_forecast.png", replace

```

Mitchell, N.M. (2020). *Data Management Using Stata: A Practical Handbook*. (2nd ed.). Stata Press.

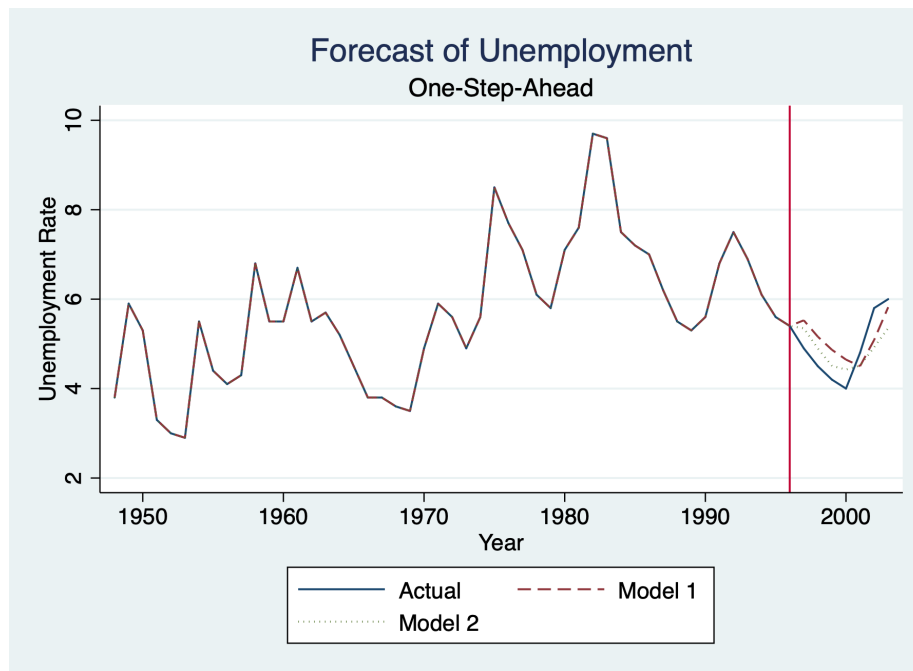


Figure 2.2: Compare Forecasts

UCLA Advanced Research Computing. (2025). *How can I access information stored after I run a command in Stata.* Retrieved from <https://stats.oarc.ucla.edu/stata/faq/how-can-i-access-information-stored-after-i-run-a-command-in-stata-returned-results/>

Wooldridge (2009). *Introductory Econometrics: A Modern Approach.* (4th ed.). South-Western Cengage Learning.